

UNITED STATES DEPARTMENT OF COMMERCE
NATIONAL BUREAU OF STANDARDS
WASHINGTON 25, D. C.

National Bureau of Standards

Test Report

2.2/171349

on

ONE MAPPING CAMERA CONE

Equipped with

C.P. Goers Planigon Lens No. 346

Submitted by

Kargl Instruments, Inc.
8123 Broadway
San Antonio 9, Texas

The lens contained in this camera has a nominal focal length of 6 inches and maximum aperture of $f/6.3$. All measurements were made at aperture $f/8$, because of limitations imposed by the apertures of the camera calibrator. All measurements were made with collimated incident light, using a K-3 filter, a tungsten source and Eastman Kodak spectroscopic emulsion Type V-F on ultra flat glass plates. Development was in D-19 at 68°F for three minutes with continuous agitation.

I. Focal Lengths

Equivalent focal length	152.19 mm
Calibrated focal length	152.18 mm

The probable errors of these determinations of focal length do not exceed ± 0.10 mm.

II. Distortion

<u>1. Distortion referred to the equivalent focal length</u>						
	<u>7.5°</u>	<u>15°</u>	<u>22.5°</u>	<u>30°</u>	<u>37.5°</u>	<u>45°</u>
Average	0	-1	-8	-17	-5	-7

<u>2. Distortion referred to the calibrated focal length</u>						
<u>Diagonal</u>	<u>7.5°</u>	<u>15°</u>	<u>22.5°</u>	<u>30°</u>	<u>37.5°</u>	<u>45°</u>
1	2	6	3	3	15	22 ✓
2	4	4	2	-5	14	23 ✓
3	-1	-3	-6	-15	0	-13
4	3	4	-8	-15	-2	1
Average	2	3	-2	-8	7	8

Radii 1 and 2 are on the same diagonal but separated in azimuth by 180°. Radii 3 and 4 are on the diagonal making an angle of 90° with 1 and 2. The values of the distortion are measured in microns and indicate the displacement of the image from its distortion-free position. A positive value indicates a displacement from the center of the plate. The probable error does not exceed ± 10 microns.

III. Resolving Power

<u>Emulsion</u>	<u>0°</u>	<u>7.5°</u>	<u>15°</u>	<u>22.5°</u>	<u>30°</u>	<u>37.5°</u>	<u>45°</u>
V-F							
Tangential	46	53	53	46	39	27	23
Radial	46	53	53	46	46	39	27
Plus-X							
Tangential	32	39	39	32	27	19	19
Radial	32	39	39	32	32	23	23

The values of the resolving power are given at 7.5° intervals from the center of the field and are obtained by photographing suitable test charts comprised of patterns of parallel lines. The series of patterns of the test chart are imaged on the negative with the lines spaced in a geometric series of the fourth root of two lines to the millimeter. The row marked "tangential" gives the number of lines per millimeter in the image on the negative of the finest pattern of the test chart that is distinctly resolved into separate lines when the lines lie perpendicular to the radius drawn from the center of the field. The row marked "radial" gives similar values for the pattern of test lines lying parallel to the radius.

IV. Principal Point of Autocollimation

The lines joining opposite pairs of collimation index markers intersect at an angle of $90^\circ \pm 1$ minute, and their intersection indicates the location of the principal point of autocollimation with a probable error not exceeding ± 0.03 mm.

V. Collimation Marker Separation

A - B 222.43 mm

C - D 222.35 mm

Markers A and B lie in the line of flight. The probable errors in these separations do not exceed ± 0.02 mm.

VI. Tangential Distortion

0°	$\pm 22.5^\circ$	$\pm 30^\circ$	$\pm 37.5^\circ$	$\pm 45^\circ$
0	4	2	1	6

The values of the tangential distortion are measured in microns and indicate the displacement of the image from its distortion-free position. These values represent a displacement of the central image from a straight line connecting corresponding image points at equal but opposite angular separations from the axis. The probable error does not exceed ± 5 microns.

The two surfaces of the filter accompanying this camera are parallel to within ten seconds of arc.

For the Director

Francis E. Washer, Chief
Refractometry Section
Metrology Division

NBS Report No. 171349
Washington, D.C.
May 28, 1962

WPTayman:ba